Factory-to-Customer Shipping Route Efficiency Analysis: A Case Study Of Nassau Candy Distributer

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***Abstract*— Nassau Candy Distributor is a large-scale distribution company with a wide range of products and services. This study focuses on assessing the efficiency of factory-to-customer shipping routes across states, regions, and shipping modes in order to close this gap. The study uses a methodical, data-driven analytical framework to assess shipping route efficiency across states and regions by utilizing shipment-level data and structured analytical techniques. The results demonstrate the existence of bottlenecks, irregular lead times, and obvious trade-offs between price and delivery time. The integration of key performance indicators like average lead time, delay rate, and route efficiency score offers practical insights to improve logistics decision-making and boost overall supply chain performance.**

Keywords*— *Supply Chain Analytics, Logistics Optimization, Route Efficiency, Shipping Lead Time, Delay Analysis, Data Analytics, Exploratory Data Analysis (EDA), Transportation Analytics, Business Intelligence, Streamlit

I. INTRODUCTION

Efficient logistics is essential to modern supply chain systems because it guarantees on-time delivery, cost containment, and customer satisfaction. Managing shipments from factories to customers gets more difficult as distribution networks spread across several regions. Despite operating on a national scale, companies like Nassau Candy Distributor frequently lack comprehensive, route-level visibility into their shipping performance, making it challenging to spot inefficiencies and delays.

The fluctuating shipping lead time, which is impacted by variables like demand distribution, transportation mode, and geographic location, is a significant problem for logistics operations. The emphasis on aggregate metrics in traditional performance evaluation techniques can mask important route-level insights. This makes it more difficult to identify delivery performance irregularities, regional bottlenecks, and underperforming routes.

This study uses a data-driven analytical framework to assess factory-to-customer shipping efficiency across states, regions, and shipping modes in order to close this gap. The study offers practical insights to improve logistics decision-making and boost overall supply chain performance by utilizing key performance indicators like average lead time, delay rate, and route efficiency score.

II. PROBLEM STATEMENT

Logistics operations produce enormous volumes of shipment and order data in large-scale distribution systems. However, when it comes to assessing performance at a fine level, this data is frequently underutilized. Businesses like Nassau Candy Distributor usually rely on aggregated metrics like total cost or delivery time, which don't give enough information about how well each factory-to-customer route is performing.

Finding inefficiencies in the logistics network is made extremely difficult by this lack of route-level visibility. High-level summaries don't show these differences, even though some routes regularly have longer lead times or delays while others operate effectively. Furthermore, geographical elements like infrastructure limitations and demand patterns unique to a given area can result in localized bottlenecks that are challenging to identify without thorough investigation.

As a result, rather than being proactive, logistics decision-making frequently becomes reactive. Only after they happen are delays addressed, which raises operating expenses and lowers customer satisfaction. Furthermore, businesses find it difficult to strike a balance between cost effectiveness and delivery speed in the absence of a methodical way to compare performance across various shipping modes.

Therefore, the lack of an organized, data-driven framework for evaluating shipping efficiency at the route level is the main issue this study attempts to address. By facilitating thorough performance evaluation, locating bottlenecks, and assisting with well-informed decision-making for optimizing logistics operations, this research seeks to close this gap.

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III. RESEARCH OBJECTIVE

This study's main goal is to create a data-driven framework for assessing the effectiveness of factory-to-customer shipping routes in a large-scale distribution network. The goal of this study is to help Nassau Candy Distributor make well-informed logistics decisions by converting shipment-level data into useful performance indicators.

The following particular goals serve as the study's compass in order to accomplish this:

1. **To analyze shipping route performance** by computing key metrics such as average lead time, delay rate, and variability across different factory-to-customer routes.
2. **To benchmark and rank routes** in order to identify the most efficient and least efficient routes within the distribution network.
3. **To examine geographic variations in performance** and identify regions or states that act as bottlenecks due to high lead times and shipment volumes.
4. **To evaluate the impact of shipping modes** on delivery performance and analyze the trade-off between cost and delivery speed.
5. **To provide actionable insights** that enable data-driven optimization of logistics operations and improve overall supply chain efficiency.
6.

IV. DATASET DESCRIPTION

Shipment-level records from Nassau Candy Distributor's logistics operations make up the dataset used in this study. It offers thorough data on order processing, shipping operations, customer locations, and financial performance, allowing for in-depth examination of the effectiveness of factory-to-customer routes.

Each observation in the dataset represents an individual shipment and includes variables related to order details, shipping characteristics, geographic information, and financial metrics. These attributes are utilized to derive key performance indicators such as shipping lead time, delay rate, and route efficiency.

Category	**Attribute**	**Description**
Order Information	Row ID	Unique identifier for each record
	Order ID	Unique identifier for each order
	Order Date	Date when the order was placed
Shipping Information	Ship Date	Date when the order was shipped
	Ship Mode	Shipping method (Standard, First Class, etc.)
Customer Information	Customer ID	Unique customer identifier
	Country/Region	Customer's country or region
	City	Customer's city
	State/Province	Customer's state or province
	Postal Code	ZIP or postal code
Geographic Segmentation	Region	Customer region classification
Product Information	Product ID	Unique product identifier
	Product Name	Name of the product
	Division	Product category/division
Financial Metrics	Sales	Total sales value of the order
	Cost	Cost associated with the product
	Gross Profit	Profit calculated as Sales − Cost
Operational Metrics	Lead Time (Derived)	Difference between Ship Date and Order Date

Table 1: Dataset Attributes Description

V. ANALYTICAL METHODOLOGY

The efficiency of factory-to-customer shipping routes is assessed in this study using a methodical, data-driven analytical framework. Through a series of clearly defined steps, such as data preprocessing, feature engineering, route-level aggregation, performance benchmarking, and multi-dimensional analysis, the methodology is intended to transform unprocessed shipment data into actionable insights. In order to accurately identify inefficiencies and optimization opportunities, the overall approach makes sure that logistics performance is analyzed at both granular and aggregated levels.

A. Data Preprocessing

Preparing the dataset to guarantee accuracy and consistency is the first step in the analysis process. This entails handling missing or inconsistent values, eliminating records with invalid or negative lead times, and validating date fields like order and ship dates. In order to preserve consistency throughout the dataset, geographic characteristics like state and region are also standardized. To guarantee the accuracy and dependability of later analysis, these preprocessing procedures are crucial.

B. Feature Engineering

In order to facilitate meaningful analysis, additional variables are derived after data preprocessing. The difference between the ship date and the order date, which indicates the overall delivery time for each shipment, is the primary performance metric, shipping lead time. Routes are defined by mapping

factory locations to customer destinations at the state and regional levels in order to make route-level evaluation easier. Furthermore, shipping modes are divided into more general categories, differentiating between options for standard and expedited delivery. The structure required to analyze performance across several dimensions is provided by these engineered features.

C. Route Aggregation and Performance Metrics

Key performance indicators are calculated by aggregating the dataset at the route level after feature engineering. These include lead time variability, average lead time, and total shipment volume, which show how consistently deliveries are made. Furthermore, the percentage of shipments that surpass a predetermined lead time threshold is used to compute the delay rate. In order to facilitate comparative analysis and performance-based route ranking, a normalized route efficiency score is also generated.

D. Efficiency Benchmarking

Routes are ranked according to the calculated efficiency metrics in order to systematically assess performance. This benchmarking procedure makes it possible to identify underperforming routes with significant delays and irregularities as well as top-performing routes with low average lead times and little variability. The analysis supports targeted optimization strategies and identifies areas that need operational improvements by creating a comparative framework.

E. Geographic Analysis

Geographic analysis is incorporated into the methodology to investigate spatial differences in logistics performance. The analysis finds regional inefficiencies and differences in delivery performance at the state level. To comprehend macro-level dynamics, broader patterns and trends are seen at the regional level. Logistics congestion or operational inefficiencies are most noticeable in bottleneck regions, which are identified by a combination of high shipment volume and above-average lead times.

F. Ship Mode Performance Analysis

Shipping performance is assessed across various modes of transportation to further improve the analysis. To determine each ship mode's relative efficiency, average lead times and delay rates are compared. The analysis also looks at the trade-off between operational cost and delivery speed, offering insights into how effective standard versus expedited shipping options are. This makes it possible to comprehend logistics performance across various delivery strategies more thoroughly.

G. Ship Mode Performance Analysis

An interactive dashboard created with Streamlit and Plotly presents the analytical results to aid in interpretation and decision-making. Customized analysis is made possible by the dashboard's ability to apply dynamic filters based on date range, region, state, and shipping mode. It offers visual aids like ship mode performance charts, geographic heatmaps, route efficiency comparisons, and thorough route-level drill-downs. This interactive implementation makes insights more accessible and makes data-driven logistics optimization easier.

VI. EXPLORATORY DATA ANALYSIS

To comprehend the underlying structure, distribution, and patterns in the shipment dataset, exploratory data analysis (EDA) was carried out. Finding patterns, irregularities, and connections between the variables that affect shipping performance is a critical function of this step. Shipment volume, lead time distribution, geographic variation, and shipping mode characteristics are the main topics of the analysis.

A. Distribution of Shipping Lead Time

To evaluate overall delivery performance and shipment variability, the distribution of shipping lead time was examined. The findings show a broad variation in delivery times, indicating irregularities in logistics processes. While most shipments are finished in a reasonable amount of time, a sizable percentage have noticeably longer lead times, which suggests that there are delays and inefficiencies. A distribution that is skewed to the right indicates that while extreme delays are uncommon, they significantly affect average performance. This emphasizes how crucial it is to minimize both average lead time and variability in order to guarantee constant service levels.

B. Distribution of Shipping Lead Time

To comprehend the distribution of demand within the network, shipment volume was analyzed across various routes, states, and regions. High-demand corridors are identified by the analysis, which shows that some routes handle a disproportionately large number of shipments. Since inefficiencies in these areas can have a substantial impact on operational outcomes, these high-volume routes are essential to overall logistics performance. Furthermore, an unequal distribution of shipment volume among regions points to possible imbalances in logistics planning and resource allocation, which could cause delays in areas with high demand.

C. Geographic Distribution of Performance

Significant differences in shipping performance between states and regions are revealed by geographic analysis. Higher average lead times are regularly seen in some areas, which may be a sign of operational inefficiencies, longer transportation distances, or infrastructure constraints. Lead time analysis and shipment volume are combined to identify specific bottleneck areas with high demand and delays. Because they directly affect the overall efficiency of delivery, these regions are crucial areas that need to be optimized.

D. Shipping Mode Analysis

To determine how various shipping methods affected delivery time and dependability, their performance was examined. According to the findings, expedited shipping methods typically result in shorter lead times than regular shipping. There is a clear trade-off between cost and efficiency, though, as this speed increase comes at a higher operating cost.

Additionally, variations in lead times between shipping modes suggest that some are more reliable than others, which is crucial for guaranteeing dependable delivery performance.

E. Identification of Delays and Outliers

Finding delayed shipments and extreme lead time values was another aspect of EDA. In order to determine the frequency of delays, a delay threshold was established to categorize shipments as delayed. According to the analysis, delays are concentrated in particular routes and areas rather than being evenly distributed. Additionally, outlier analysis reveals the existence of extreme cases with abnormally long lead times, which can be linked to geographical difficulties, operational disruptions, or inefficiencies in particular routes. In order to improve system performance, these outliers must be addressed because they significantly increase overall variability.

VII. KEY PERFORMANCE INDICATOR

A set of key performance indicators (KPIs) is established in order to assess shipping route efficiency quantitatively. These metrics offer a uniform framework for gauging the effectiveness of logistics across shipping modes, regions, and routes. Delivery speed, consistency, frequency of delays, and operational scale are just a few of the performance dimensions captured by the chosen KPIs.

A. Shipping Lead Time

The entire time needed to deliver an order, from order placement to shipment completion, is known as the shipping lead time. It is the main measure of delivery effectiveness.

Lead Time = Ship Date – Order Date

A lower lead time shows faster delivery and improved logistics performance.

B. Average Lead Time

An overview of shipping efficiency is provided by average lead time, which calculates the average delivery time for a particular route or segment.

$$\textbf{Average Lead Time} = \frac{\sum \boldsymbol{Lead\ Time}}{\boldsymbol{N}}$$

where N represents the total number of shipments.

C. Delay Rate

The percentage of shipments that surpass a predetermined lead time threshold is measured by the delay rate. It shows how dependable the logistics system is.

$$\boldsymbol{Delay\ Rate} = \frac{\boldsymbol{Number\ of\ Delayed\ Shipments}}{\boldsymbol{Total\ Shipments}}$$

Poor delivery dependability and possible operational inefficiencies are indicated by a higher delay rate.

D. Lead Time Variability

The standard deviation of lead times is used to calculate lead time variability, which quantifies the consistency of delivery performance.

$$\sigma = \sqrt{\frac{1}{N}\sum(x_i - \mu)^2}$$

where xi stands for individual shipment lead times and μ\mu is the average lead time. More consistent delivery performance is indicated by lower variability.

E. Route Volume

The total number of shipments handled by a particular factory-to-customer route is known as the route volume. It displays the load and operational significance of each route.

Route Volume = Total Number of Shipments per Route

High-volume routes need to be closely monitored because they are essential to the overall performance of the system.

F. Route Efficiency Score

A normalized metric for comparing performance across routes with various characteristics is the route efficiency score. Lead time values are scaled in relation to the observed range in order to derive it.

$$Efficiency\ Score = 1 - \frac{Lead\ Time - Min}{Max - Min}$$

A higher efficiency score shows better route performance.

VIII. DASHBOARD IMPLEMENTATION

Using Streamlit and Plotly, a web-based dashboard was created to convert analytical insights into an interactive and user-friendly system. Through dynamic filtering and visualization, the dashboard allows for real-time investigation of shipping performance across routes, regions, and shipping modes.

The following link will take you to the interactive dashboard created for this project:
Dashboard Link: *[Nassau Candy – Route Efficiency · Streamlit]*

A. Route Efficiency Overview

Through key performance indicators like total shipments, average lead time, delay rate, and profitability, the dashboard offers a thorough overview of route-level performance. Efficient and inefficient routes can be quickly identified thanks to a bar chart visualization that shows differences in average lead times across various factory-to-region routes.

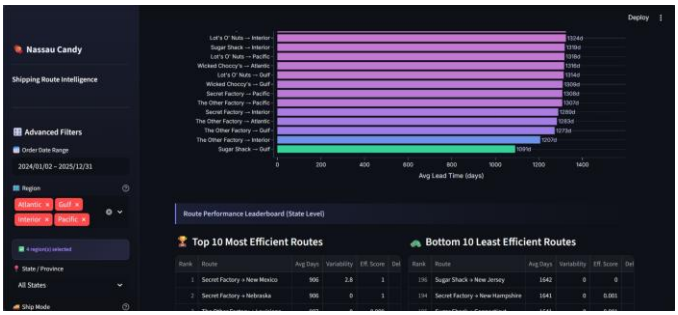

Fig 1. Route Efficiency Overview

The dashboard presents route-wise average lead time, and top 10 and bottom 10 routes,and highlighting significant variation in delivery performance across regions.

B. Geographic Visualization

To examine spatial differences in shipping efficiency between various states, a geographic heatmap is integrated. Geographic bottlenecks can be found by using this visualization to identify areas with higher lead times and delay rates.

Fig 2. Geographic Heatmap of Shipping Performance

The heatmap illustrates state-level variations in shipping efficiency, highlighting regions with higher delivery delays.

C. Ship Mode Analysis

A specific area of the dashboard is devoted to comparing the effectiveness of various shipping methods. Differences in lead time and delay rates between shipping methods are examined using visualizations like distribution plots and bar charts.

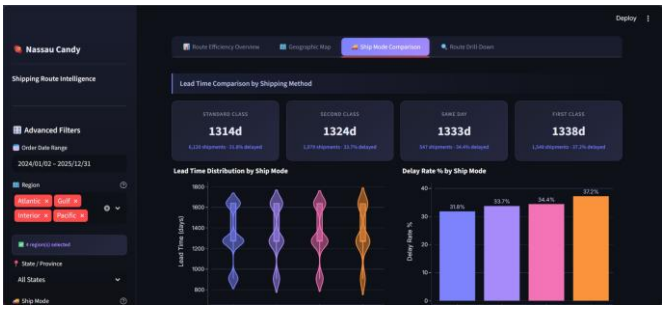

Fig 3. Ship Mode Performance Comparison

The comparison shows that expedited shipping modes achieve lower lead times but involve higher operational costs.

D. Route Drill-Down Analysis

A comprehensive drill-down module enables users to thoroughly examine specific routes. This allows for a detailed examination of performance and includes shipment-level schedules, lead time distributions, and ship mode breakdowns.

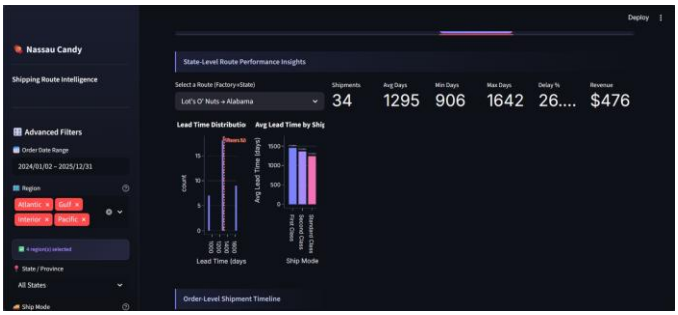

Fig 4. Route Drill-Down Analysis

The drill-down view provides detailed insights into shipment patterns and variability for a selected route.

E. System Functionality and User Interaction

Users can personalize their analysis with the dashboard's interactive filtering based on date range, region, state, and shipping mode. Real-time filtering and dynamic visualizations are integrated to improve usability and facilitate effective logistics performance exploration.

IX. KEY INSIGHTS

The route-level evaluation and exploratory analysis offer several important insights into Nassau Candy Distributor's logistics performance. These observations draw attention to important inefficiencies, performance trends, and distribution network optimization opportunities.

A. Significant Variation in Route Performance

Significant differences in shipping performance between factory-to-customer routes are revealed by the analysis. Certain routes consistently show low lead times and high efficiency, while others have longer delivery times and more delays. This suggests that route-specific optimization techniques are necessary because logistics performance is not consistent.

B. Presence of High-Impact Bottleneck Regions

A combination of high shipment volume and above-average lead time identifies some areas as bottlenecks. These areas have a disproportionate effect on the overall performance of the system, indicating that focused enhancements in these areas can result in notable increases in efficiency.

C. Trade-off Between Cost and Delivery Speed

A clear trade-off between delivery speed and operating cost is highlighted by the comparison of shipping modes. While standard shipping is more affordable but takes longer, expedited shipping methods result in quicker delivery times but are more expensive. This highlights the necessity of a strategic and well-rounded shipping approach.

D. High Variability in Delivery Performance

High lead time variability across multiple routes is a sign of inconsistent logistics operations. It is challenging to guarantee constant service levels because of this variability, which diminishes predictability and dependability. Therefore, addressing variability is just as crucial as cutting the average delivery time.

E. Concentration of Delays in Specific Routes

Delays are concentrated in particular routes and areas rather than being dispersed randomly. This implies that inefficiencies are localized rather than systemic, enabling focused interventions to boost performance without completely redesigning the network.

F. Importance of Data-Driven Monitoring

The value of real-time monitoring in logistics operations is illustrated through the use of an interactive dashboard created with Streamlit. Making better decisions and proactively identifying problems are made possible by the ongoing monitoring of key performance indicators.

X. STRATAGIC RECOMMENDATION

Several strategic recommendations are put forth to improve logistics efficiency and lower operational inefficiencies within Nassau Candy Distributor based on the thorough analysis of shipping routes, geographic performance, and shipping modes. These suggestions are meant to facilitate data-driven decision-making, reduce delays, and enhance delivery performance.

A. Optimization of High-Delay Routes

Routes that have been found to have persistently high lead times and delay rates ought to be given priority for optimization. This could entail adjusting routing plans, enhancing factory-distribution center coordination, or shifting resources to routes with high demand. Network efficiency can be greatly increased by concentrating on underperforming routes.

B. Bottleneck Mitigation in High-Volume Regions

Critical bottlenecks should be identified in areas with high shipment volumes and above-average lead times. Congestion can be lessened and delivery times can be improved with targeted interventions like building regional distribution hubs, optimizing delivery schedules, and expanding transportation capacity.

C. Strategic Use of Shipping Modes

A clear trade-off between delivery speed and cost across shipping modes is highlighted by the analysis. A hybrid shipping approach is advised, in which standard shipping is kept for deliveries that are cost-sensitive and expedited shipping is only used for high-priority or delay-prone routes. This strategy guarantees a balance between cost containment and operational effectiveness.

D. Reduction of Lead Time Variability

Inconsistent delivery performance is indicated by high lead time variability. Standardizing logistics processes, increasing coordination throughout the supply chain, and improving scheduling accuracy should all be priorities. For delivery results to be dependable and predictable, variability must be reduced.

E. Data-Driven Decision Support

Streamlit's interactive dashboard implementation makes it possible to continuously monitor logistics performance. It is advised to incorporate these tools into routine operational procedures so that decision-makers can recognize problems instantly and take preventative action.

F. Continuous Performance Monitoring

It is crucial to set up a system for ongoing tracking of key performance metrics like lead time, delay rate, and route efficiency score. Frequent assessment of these metrics will aid in monitoring advancements, spotting new problems, and guaranteeing ongoing logistics operations optimization.

XI. CONCLUSION

This study offers a thorough, data-driven analysis of Nassau Candy Distributor's factory-to-customer shipping route efficiency. The study finds notable differences in delivery performance between routes, regions, and shipping modes by utilizing shipment-level data and structured analytical techniques. The results demonstrate the existence of bottlenecks, irregular lead times, and obvious trade-offs between price and delivery time.

A methodical assessment of logistics performance is made possible by the integration of key performance indicators like average lead time, delay rate, and route efficiency score. Using an interactive dashboard also makes it easier to monitor in real time and makes complex logistics data easier to understand, which promotes proactive and well-informed decision-making.

All things considered, the study shows how well data analytics can convert unprocessed operational data into useful insights. Organizations can increase overall supply chain efficiency, optimize resource allocation, and improve delivery reliability by implementing a route-level analytical approach. The significance of switching from reactive to data-driven logistics management in contemporary distribution systems is emphasized by this work.

XII. REFERENCES

[1] Unified Mentor Pvt. Ltd., "Project Guidelines: Factory-to-Customer Shipping Route Efficiency Analysis for Nassau Candy Distributor," Internship Project Portal, Nagpur, India, 2026.

[2] **Nassau Candy Distributor**, "Order and Shipment Logistics Dataset (2024-2026)," provided via Unified Mentor Internship Portal. [Online]. Available: [Nassau Candy Distributor.csv - Google Drive].

[3] **S. Ambade**, "Shipping Route Efficiency Analysis for Nassau Candy Distributor: Exploratory Data Analysis and Feature Engineering," Jupyter Notebook documentation, Mar. 2026.

[4] **W. McKinney**, "Data Structures for Statistical Computing in Python," in *Proceedings of the 9th Python in Science Conference*, Austin, TX, 2010, pp. 51-56.

[5] **Streamlit Inc.**, "Streamlit Documentation: Building Interactive Data Applications for Logistics," 2026. [Online]. Available: https://docs.streamlit.io.

[6] **J. Hunter**, "Matplotlib: A 2D Graphics Environment," *Computing in Science & Engineering*, vol. 9, no. 3, pp. 90-95, 2007.

[7] **M. Waskom et al.**, "seaborn: statistical data visualization," *Journal of Open Source Software*, vol. 6, no. 60, p. 3021, 2021.

[8] **Nassau Candy Distributor**, "Internal Logistics Resource: Factories Coordinates and Product Correlation Matrix," Technical Reference Guide, 2026.